Lec-9
31/01/26 4:41 PM

Sorting → ↑
↳ Arrang ↑ ↓

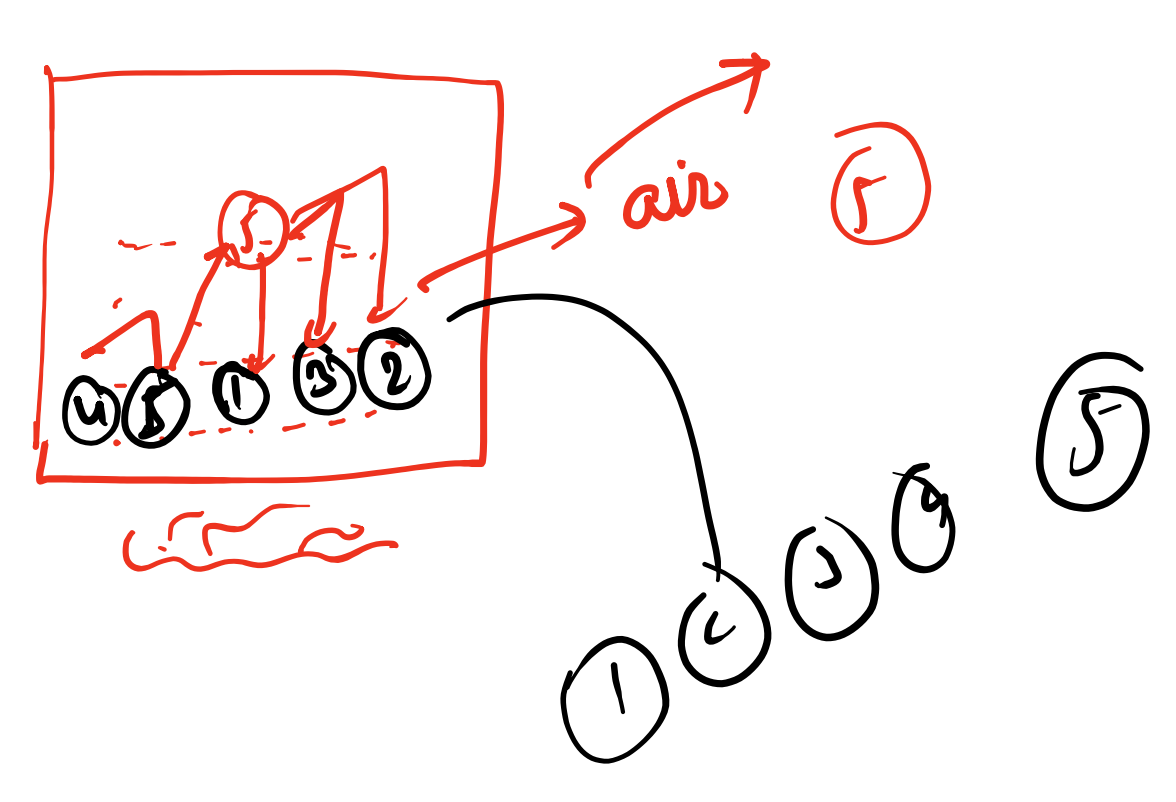
(1 to n)
n-1
n-2

Queue

9:15

① bubble
(ii) selection
(iii) Insertion
(iv) merge
(v) quick
(vi) heap
(vii) counting
(viii) D.N.F

bubble Sort



① [4, 5, 3, 2, 1]

i=0 [4, 5, 3, 2, 1]

i=1 [4, 3, 5, 2, 1]

i=2 [4, 3, 2, 5, 1]

[4, 3, 2, 1, 5]

(ii) [4, 3, 2, 1, 5]
[3, 4, 2, 1, 5]
[3, 2, 4, 1, 5]
[3, 2, 1, 4, 5]

(iii) [3, 2, 1, 4, 5]
[2, 3, 1, 4, 5]
[2, 1, 3, 4, 5]

(iv) [2, 1, 3, 4, 5]
[1, 2, 3, 4, 5]

```
for( turn=1 turn<n, turn++) {
    for( i=0   i<[          ]   i++) {
        if (arr(i) > arr(i+1)) {
            (i, i+1) swap
        }
    }
}
```

arr.lem - turn =
5-1=4 | 5-2=3
5-3=2
5-4=1

i=0 4 5 3 2 1
i=1 4 5 3 2 1
4 3 5 2 1
i=2 4 3 2 5 1
i=3 4 3 2 1 5

(ii) [4, 3, 2, 1, 5]
i=0 3, 4, 2, 1, 5
i=1 3 2 4 1 5
i=2 3 2 1 4 5

(iii) [3, 2, 1, 4, 5]
i=0 2, 3, 1, 4, 5
i=1 2 1 3 4 5

(iv) 2 1 3 4 5
i=0 1 2 3 4 5

```
int[] arr = { 4, -1, 5, 3, 2, 1, 7 };
```
0 1 2 3 4 5 6

mini -> idx -> 2 3 4 5

i=3, arr(2) > arr(3)
i=4 arr(3) > arr(4)
i=5 arr(4) > arr(5)
i=6 arr(5) > arr(6)

```
mini=idx
for( i=idx+1  i<n  i++ {
    if (arr[mini] > arr(i)) {
        mini=i
    }
}
```

```java
public static int min_form_ith_index(int[] arr, int idx) {
    int mini = idx;
    for (int i = idx+1; i < arr.length; i++) {
        if (arr[mini] > arr[i]) {
            mini = i;
        }

    }
    return mini;
}
```

```java
int[] arr = { 4, -1, 5, 3, 2, 1, 7 };
```
0 1 2 3 4 5 6

i=0 [-1, 4, 5, 3, 2, 1, 7] → 5
i=1 [-1, 1, 5, 3, 2, 4, 7] → 4
i=2 [-1, 1, 2, 3, 5, 4, 7] → 3
i=3 [-1, 1, 2, 3, 5, 4, 7] → 5
i=4 [-1, 1, 2, 3, 4, 5, 7]
i=5 [-1, 1, 2, 3, 4, 5, 7]
i=6 [-1, 1, 2, 3, 4, 5, 7]

```java
int[] arr = { 1, 4, 7, 8, 9, 11, 2 };
```

j=5 1 4 7 8 9 2 11
j=4 1 4 7 8 2 9 11
j=3 1 4 7 2 8 9 11
j=2 1 4 2 7 8 9 11
j=1 1 2 4 7 8 9 11
j=0

1 4 7 8 9 2 11
1 4 7 8 2 9 11
1 4 7 2 8 9 11
1 4 2 7 8 9 11
1 2 4 7 8 9 11

```java
int[] arr = { 1, 4, 7, 8, 9, 11, 2 };
```
0 1 2 3 4 5 6

i=6
item=arr(i)
j=i-1
5

```
while( j>=0 && arr[j] > item) {
    arr[j+1] = arr(j)
    arr(j) = item
    j--
}
```
j+1

1 4 7 8 9 10 11

0 1 2 3
[2, 3, 4, 1]
i=3
item=1
j=i-1=2
j=2
j=1 arr(2)>1
arr(1)>1
j=0 arr(0)>1
j=-1
[2, 3, 1, 4]
[2, 1, 3, 4]
[1, 2, 3, 4]

```java
int[] arr = { 9, 1, 7, 4, 8, 2, 11 };
```

[9, 1, 7, 4, 8, 2, 11]
[1, 9, 7, 4, 8, 2, 11]
[1, 7, 9, 4, 8, 2, 11]
[1, 4, 7, 9, 8, 2, 11]
[1, 4, 7, 8, 9, 2, 11]
[1, 2, 4, 7, 8, 9, 11]

[7]
[9 1] i=1
[1 9 7] i=2
[1 7 9 4] i=3
[1, 4, 7, 9, 8] i=4
[1, 4, 7, 8, 9, 2] i=5
[1, 2, 4, 7, 8, 9, 11] i=6

ans → ~~2~~ ~~5~~ 6

0 1 2 3 4 5
[2, 3, -7, 4, -1, 3]

[0-0] 2
[0-1] 2 3 → 5
[0-2] 2 3 -7 → -2
[0-3] 2 3 -7 4 → 2
[0-4] 2 3 -7 4 -1 → 1
[0-5] 2 3 -7 4 -1 3 → 4

[1-1] 3
[1-2] 3 -7 → -4
[1-3] 3 -7 4 → 0
[1-4] 3 -7 4 -1 → -1
[1-5] 3 -7 4 -1 3 ②

[2-2] -7
[2-3] -7 4 -3
[2-4] -7 4 -1 -4
[2-5] -7 4 -1 3 -1

[3-3] 4
[3-4] 4 -1 3
[3-5] 4 -1 3 6

[4-4] -1
[4-5] -1 3

[5-5] 3

```
ans = -∞
for( i=0   i<n   i++) {
    int sum=0
    for( j=i   j<n   j++) {
        sum = sum + arr(j)
        if (ans < sum) {
            ans=sum
            p=i, q=j
        }
    }
}
```

p=0
q=0

0 1 2 3 4 5 6 7 8 9
[2, 3, -7, 4, -1, 3, -9, 3, -1, 2]
-2
-3

```java
public static int SubArray_Sum(int[] arr) {
    int ans = Integer.MIN_VALUE;// -2^31
    int curr_sum = 0;
    for (int i = 0; i < arr.length; i++) {
        curr_sum += arr[i];
        ~~ans = Math.max(ans, curr_sum);~~
        if (curr_sum < 0) {
            curr_sum = 0
        }
    }
    return ans;
}
```

if come < ans {

ans = ~~2~~ ~~5~~ 6

cs = 2 + 3 = 5 - 7
= -2   0 + 4 = 4 - 1
= 3 + 3
= 6 - 9
= -3   0 + 3 - 1
= 2 + 2 = 4

p   q → (-1)

2 → 2
2 3 → 5
2 3 -7 → -2
2 3 -7 4 → 2
2 3 -7 4 -1 → 1
2 3 -7 4 -1 3 → 4
2 3 -7 4 -1 3 -9 → -5
2 3 -7 4 -1 3 -9 3 → -2
2 3 -7 4 -1 3 -9 3 -1 → -3
2 3 -7 4 -1 3 -9 3 -1 2 → -1

0 1 2 3 4 5 6 7 8
[2 7 3 7 3 7 7 5 7]

⑦

x _ x _ x _ x _ x _ x _ x
0 1 2 3 4 5 6 7 8 9 10 11 12

c = 2 7 8 7
vote = 1 ✓
0 1 2